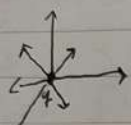


Exercise 3

i) i)



$E(r)$ is a vector value that is represented by electric field lines from the point charge.

The field lines show "how much" and what direction a positive particle would be affected. The "density of field" lines gives a visualisation of the strength of the E-field and can be seen as the field lines become longer, the density decreases and the strength drops. This is the same as calculating the surface of a sphere with radius r which gives the $1/r^2$ (and $1/4\pi$) dependence.

ii) E is again the electric field represented as a vector pointing in/out of its source. da represents a surface area, and so the dot product $E \cdot da$ gives how "many" of the field lines pass through the area (or how much of them). Flux is useful as it's often used in combination with Gauss's Law to find how much a charge's electric field affects other areas/surfaces.

iii) 1. If a charge is enclosed inside a surface, then the electric lines in/out of the charge will all pass through the surface, giving a net electric flux.

2. If a charge is outside an enclosed surface then the electric field lines that enter the surface will also exit the surface on the other side, giving the sum of e-field lines leaving and entering being 0, and thus no flux.

iv) Given a surface charges Q_{enc} in a sphere with radius R :

$$\oint E \cdot da = \frac{1}{4\pi\epsilon_0} \int \int \left(\frac{Q_{enc}}{r^2} \right) \cdot (r^2 \sin\theta d\theta d\phi \hat{r})$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi Q_{enc} \sin\theta d\theta d\phi$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} 2Q_{enc} d\phi = \frac{Q_{enc}}{\epsilon_0}$$

This is a constant because the surface area of the electric-field scales with r^2 and the E-field strength drops with $1/r^2$. This makes the flux independent of radius.

$$v) \oint \mathbf{E} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{E}) d\tau = \frac{Q_{enc}}{\epsilon_0}$$

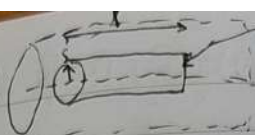
If we rewrite Q_{enc} in terms of charge density ρ :

$$Q_{enc} = \int_V \rho d\tau \Rightarrow \int_V (\nabla \cdot \mathbf{E}) d\tau = \int_V \frac{\rho}{\epsilon_0} d\tau$$

And since this goes for any volume the integrals should be equal:

$$\Rightarrow \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

iv) The equations $\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$ & $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ shows that the total flux through a surface is constant given its enclosed in it, and that the charge density and the rate of flux for a given E-field is also a constant. Both involving dividing by permittivity.

2) i)  $\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc}$

where $Q_{enc} = \int_V \rho(s') d\tau = \int_0^{2\pi} \int_0^l \int_0^s (k s') \cdot (s' ds' d\theta dz)$
 $\Rightarrow 2\pi k l \int_0^s s' ds' = \frac{2}{3} \pi k l s^3$ so:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{2}{3} \pi k l s^3$$

using symmetry and that the magnitude of $\mathbf{E}$ is constant over gaussian fields and the ends contribute nothing:

$$\Rightarrow |\mathbf{E}| \int_0^{2\pi} \int_0^l s d\theta dz = 2\pi l s |\mathbf{E}|$$

$$\Rightarrow |\mathbf{E}| \pi s l = \frac{2}{3} \pi k l s^3$$

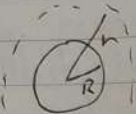
$$\Rightarrow \underline{E(s) = \frac{2}{3} k s^2 \hat{s}}$$

ii) Since $\rho=0$ for $s>a$, the Q_{enc} would be $= \frac{2}{3} \pi k l a^3$, and

$$\oint \mathbf{E} \cdot d\mathbf{a} = |\mathbf{E}| \int_0^{2\pi} \int_0^l s d\theta dz = 2\pi l s |\mathbf{E}| \quad |s>a$$

$$\Rightarrow \underline{E(s) = \frac{1}{3} k \frac{a^3}{s} \hat{s}} \quad \text{iii) if } s=a \text{ they are the same.}$$

3) a)  $\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0} = 0 \quad r < R$

 $\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$ where

$Q_{enc} = \int \sigma da = \int_0^{2\pi} \int_0^{\pi} \sigma R^2 \sin\theta d\theta d\phi$

$\Rightarrow Q_{enc} = 4\pi R^2 \sigma$

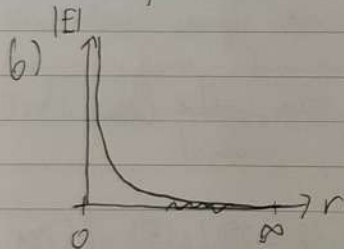
and

$\oint \mathbf{E} \cdot d\mathbf{a} = |\mathbf{E}| \oint da = |\mathbf{E}| \int_0^{2\pi} \int_0^{\pi} r^2 \sin\theta d\theta d\phi$

$\Rightarrow |\mathbf{E}| 4\pi r^2$ so that

$|\mathbf{E}| 4\pi r^2 = \frac{4}{\epsilon_0} \pi R^2 \sigma \Rightarrow \underline{\underline{E(r) = \frac{R^2 \sigma}{\epsilon_0 r^2} \hat{r}}}$

$\mathbf{E}$ will point radially outward.



Q) If $r < R$ then $Q_{enc} = \int \rho d\tau = \int_0^{2\pi} \int_0^{\pi} \int_0^r \rho r'^2 \sin\theta d\theta d\phi dr'$

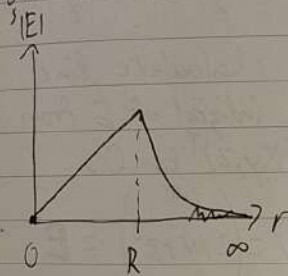
$\Rightarrow Q_{enc} = \frac{4}{3} \pi r^3 \rho$ and then

$\oint \mathbf{E} \cdot d\mathbf{a} = |\mathbf{E}| \int_0^{2\pi} \int_0^{\pi} r^2 \sin\theta d\theta d\phi = |\mathbf{E}| 4\pi r^2$ so that

$\Rightarrow |\mathbf{E}| 4\pi r^2 = \frac{4}{\epsilon_0} \pi r^3 \rho \Rightarrow \underline{\underline{E(r) = \frac{\rho r}{3\epsilon_0} \hat{r} \text{ for } r < R}}$

If $r > R$ then $Q_{enc} = \frac{4}{3} \pi R^3 \rho$ and then

$\oint \mathbf{E} \cdot d\mathbf{a} = |\mathbf{E}| 4\pi r^2 = \frac{4}{\epsilon_0} \pi R^3 \rho \Rightarrow \underline{\underline{E(r) = \frac{R^3 \rho}{3\epsilon_0 r^2} \hat{r} \text{ for } r > R}}$



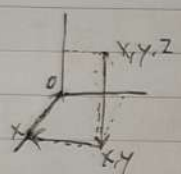
$$4) i) \nabla \times E = 0 \Rightarrow \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{x} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{y} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{z} = 0$$

$$a) (0 - 2y) \hat{x} + (0 - 3z) \hat{y} + (0 - x) \hat{z} \neq 0$$

$$b) (2z - 2z) \hat{x} + (0 - 0) \hat{y} + (2y - 2y) \hat{z} = 0$$

Since b) has no curl, it is a possible E-field.

$$ii) \begin{pmatrix} y^2 \\ 2xy + z^2 \\ 2yz \end{pmatrix} V(x, y, z) = \int_x^0 0 dx' + \int_y^0 2xy' dy' + \int_z^0 2yz' dz'$$



$$\Rightarrow -xy^2 - yz^2$$

Calculate line integral of E from $(x, y, z)^T$ to 0.

$$iii) -\nabla V = - \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot (-xy^2 - yz^2) = \begin{pmatrix} y^2 \\ 2xy + z^2 \\ 2yz \end{pmatrix} = E$$

$$5) a) V_{ring}(z) = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma}{z} \cdot r' d\theta dr' = \frac{\sigma R^2}{2\epsilon_0 z}$$

$$b) \begin{array}{c} \theta \\ \nearrow \\ R \\ \searrow \\ z \end{array} \quad r^2 = z^2 + R^2 \quad \cos\theta = \frac{z}{\sqrt{z^2 + R^2}}$$

$$V(z) = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma_{ring}}{r} + \cos\theta dl' = \frac{\sigma_{ring}}{2\epsilon_0} \frac{z}{(z^2 + R^2)}$$

$$c) V_{total} = \frac{\sigma R^2}{2\epsilon_0 z} + \frac{\sigma_{ring} z}{2\epsilon_0 (z^2 + R^2)} = \frac{1}{2\epsilon_0} \left(\frac{\sigma R^2}{z} + \frac{\sigma_{ring} z}{(z^2 + R^2)} \right)$$

$$d) E(z) = -\nabla V = -\frac{1}{2\epsilon_0} \left(\frac{\sigma_{ring} (R^2 - z^2)}{(z^2 + R^2)^2} - \frac{\sigma R^2}{z^2} \right) \hat{z}$$